

FreeFlume — Playlist Pagination Fix

Restoring access to playlist pages beyond item 200, anonymously · June 2026

1. The symptom

Navigating a long YouTube playlist (e.g. `list=PLEPGYRPuQFbpiH53ip5dx_cbS7Bu-DWmF`, 1,005 items) showed **"No videos found"** from page 6 onward. Earlier pages worked. The same playlist was fully viewable, logged-out, in a normal web browser — so the content was public and the limitation was on *our* extraction path, not on YouTube's side.

2. Root cause

FreeFlume paginated playlists through yt-dlp:

```
yt-dlp --flat-playlist --playlist-start 101 --playlist-end 120 <playlist-url>
```

yt-dlp extracts an anonymous playlist by *walking* the list one InnerTube page at a time, following each response's `continuationToken`. For a scripted (non-browser) client, **YouTube stops emitting those continuation tokens after ~200 items**. So any page starting past item 200 came back empty → "No videos found." The page 6 boundary (item 101+) was the first to fall off the cliff once a related extraction quirk capped the first hop at 100.

Dead ends ruled out (empirically): none of these lift the cap — `youtubetab:skip=webpage` (only raises 100 → 200), visitor data, consent / warmed session cookies, a valid BotGuard PO token, or switching InnerTube client. The browser and NewPipe get further via anti-bot reputation heuristics, not a different API.

3. The insight

A YouTube playlist continuation token is not opaque — it is a **base64url-encoded protobuf**. Decoding a real one revealed it simply encodes the **playlist id and an item offset**:

```
#80226972
#2  "VLPLEPGYRPuQFbpiH53ip5dx_cbS7Bu-DWmF"    (VL + playlist id)
#3  <wrapper> → PT: { #1 varint: 100,           ← the OFFSET
                      #4 "1B9F88AE53EE5749" } ← boundary marker (optional)
#35  "PLEPGYRPuQFbpiH53ip5dx_cbS7Bu-DWmF"
```

The offset is a plain number, and the sibling boundary marker turned out to be **optional**. That means we can **craft a token for any offset and jump straight to that page in a single request** — instead of walking the chain and hitting the wall. This is the same mechanism the browser and NewPipe use internally; the difference is we synthesise the offset rather than discover it.

4. Empirical verification

Crafted offset	Result	Conclusion
20	First item returned = playlist item 21	Exact alignment: <code>offset = (page-1) · pageSize</code>
200	100 items, no marker needed	Marker is optional
1000 (page 51)	Final items, correct titles / channel / durations	Past the 200 wall ✓

The C++ token encoder was checked to reproduce a genuine YouTube token **byte-for-byte** before any live request was trusted.

5. The fix

Playlists now bypass yt-dlp and call YouTube's InnerTube `browse` API directly over Qt Network (already linked). In `src/extractor.cpp`:

- `fetchChannel()` — for `list= / /playlist` URLs, routes to `fetchPlaylistPage()`. Page 1 uses `browseId = "VL"+id` (this response also carries the total count, e.g. `"1,005 videos"`, for the pager); deeper pages POST a crafted continuation token for `offset = (page-1) · pageSize`.
- `fetchPlaylistItems()` (the play-queue) — walks offsets 0, 100, 200, ... so a queued playlist is no longer truncated at 200 either.
- Items are parsed from the modern `lockupViewModel` node: `contentId`, `title`, `channel`, and the duration badge. Token crafting is hand-rolled protobuf, mirroring the existing `buildSpParam()` approach.
- `cancel()` / `busy()` updated to track the in-flight network request alongside the yt-dlp subprocess.

How an offset token is built (C++)

```
PT = field1(varint = offset)
PTb = "PT:" + base64(PT, no padding)
wrap = { field1 = 1, field15 = PTb }; wrapB = base64(wrap), '=' → "%3D"
body = { field2 = "VL"+id, field3 = wrapB, field35 = id }
token = base64url( field80226972 = body )
```

6. Why this respects the project constraints

- **Fully anonymous** — no account, cookies, browser-cookie extraction, visitor data, or PO token. Nothing is signed in.
- **No external daemon** — unlike the shelved bgutil PO-token route; this is pure in-process C++/Qt.

- **Native** — consistent with FreeFlume's no-Electron / no-JVM stance.

7. Trade-off to watch

We now own an InnerTube `LookupViewModel1` parser. yt-dlp previously shielded us from YouTube's response-format churn; if YouTube renames those fields, playlist parsing is the first thing that breaks. The parser is deliberately minimal and defensive — `contentId` + title are required, everything else is best-effort — to limit that exposure.

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